MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

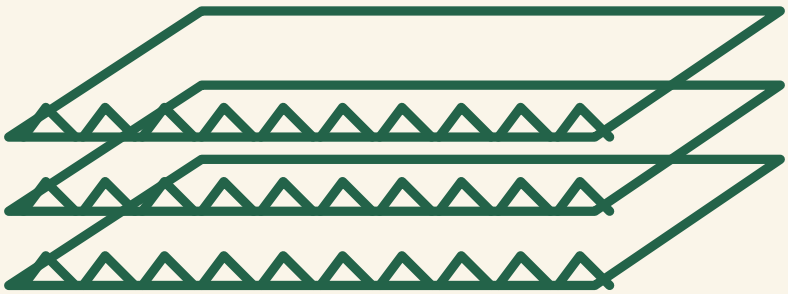
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M01

BUILD WITH...

Corrugated cardboard



Use the ridged layers of a shipping box as your main building material. Cut, fold or laminate them.

FOLD / STACK / SLOT

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

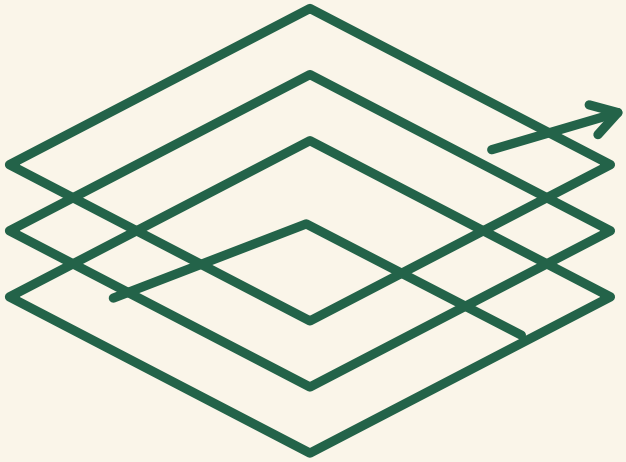
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M02

BUILD WITH...

Paper



Make paper do the work. Fold, roll, pleat or layer ordinary sheets into a useful structure.

FOLD / ROLL / PLEAT

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

MATERIALS

18 CARDS / DRAW ONE

MATERIALS M03

BUILD WITH...

Scrap wood



Give a wooden offcut another job. Make it a useful part of your contraption.

JOIN / STACK / SHAPE

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

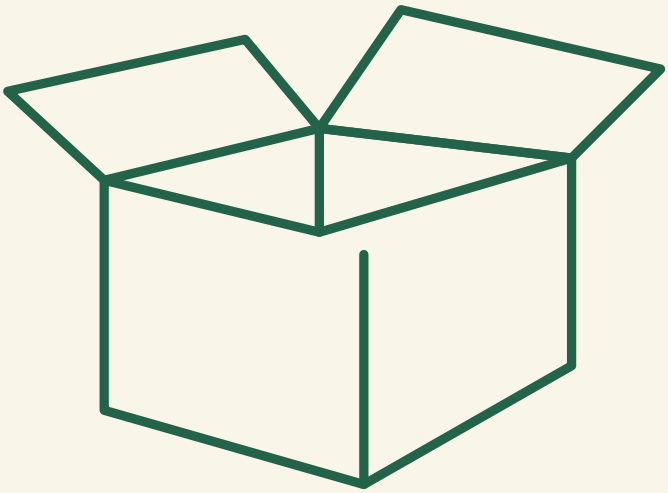
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M04

BUILD WITH...

Discarded packaging



Turn a clean box, carton or packing insert into the main structure of your build.

RESCUE / REBUILD

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

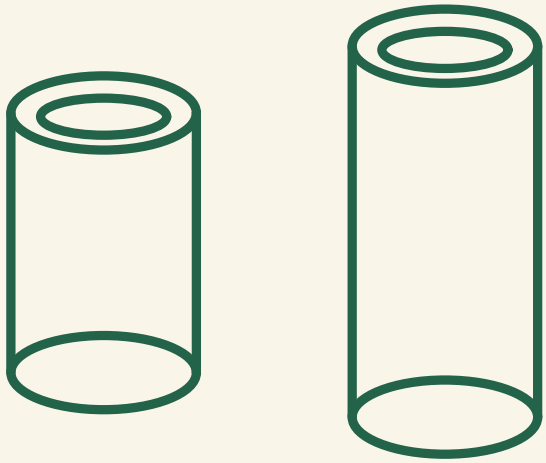
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M05

BUILD WITH...

Paperboard tubes



Build around one or more paperboard tubes. Use their hollow shape, or cut and reshape them.

ROLL / NEST / JOIN

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

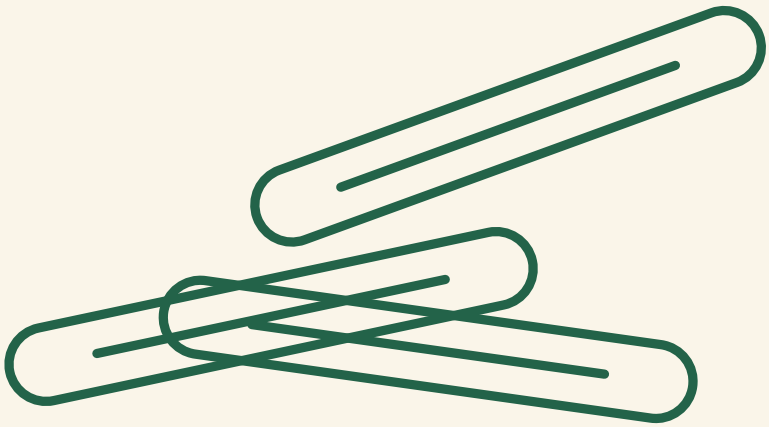
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M06

BUILD WITH...

Craft sticks



Use craft sticks as the main structural pieces. Connect them into something that works.

BRIDGE / LINK / LAYER

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

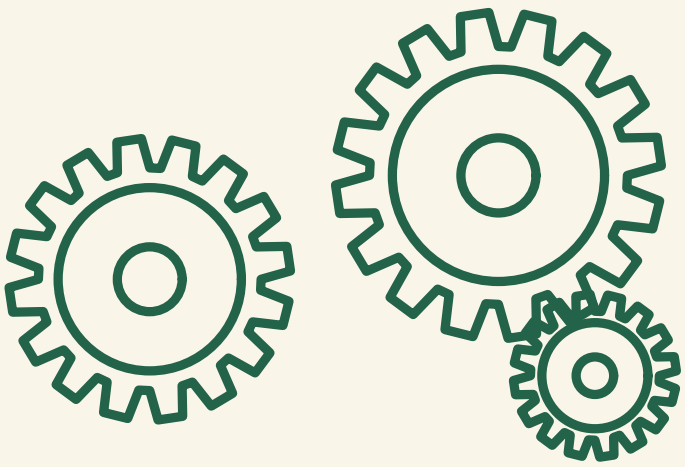
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M07

BUILD WITH...

Bottle caps



Make clean bottle caps an essential part of your build. Explore their rims, circles and small size.

ROLL / STACK / SORT

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

MATERIALS

18 CARDS / DRAW ONE

MATERIALS M08

BUILD WITH...

Plastic bottles



Give a clean plastic bottle a new purpose. Use its body, neck or flexible walls in your build.

SQUEEZE / SHAPE / HOLD

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

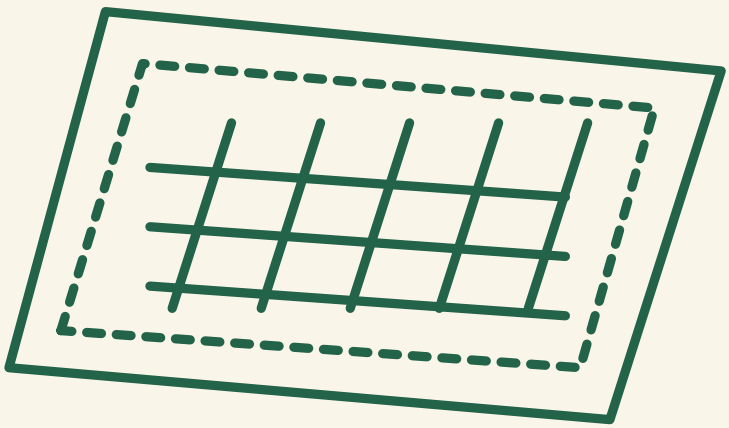
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M09

BUILD WITH...

Fabric scraps



Make fabric an essential working part. Fold, stretch, tie or stitch your scraps into shape.

STITCH / STRETCH / FOLD

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

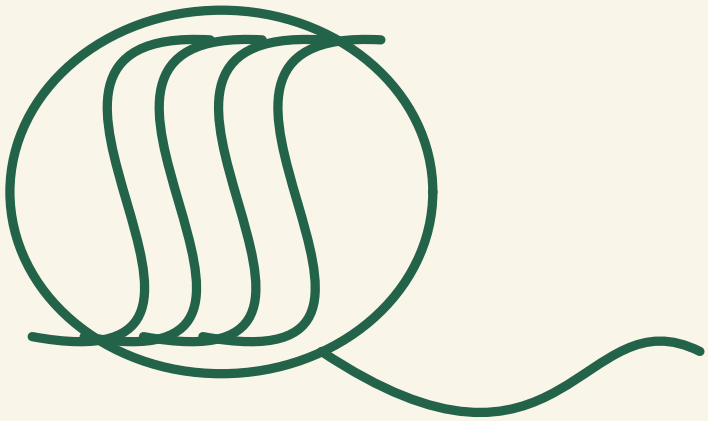
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M10

BUILD WITH...

String or yarn



Use string or yarn to carry a load, connect parts or transmit motion. Put tension to work.

TIE / WEAVE / PULL

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

MATERIALS

18 CARDS / DRAW ONE

MATERIALS M11

BUILD WITH...

Thin craft wire



Bend thin craft wire into a useful structure or moving part. Give those lines a job to do.

BEND / LOOP / TWIST

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

MATERIALS

18 CARDS / DRAW ONE

MATERIALS M12

BUILD WITH...

Rubber bands



Put rubber bands to work. Use their stretch, grip or stored energy as a key part of your build.

STRETCH / GRIP / RETURN

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

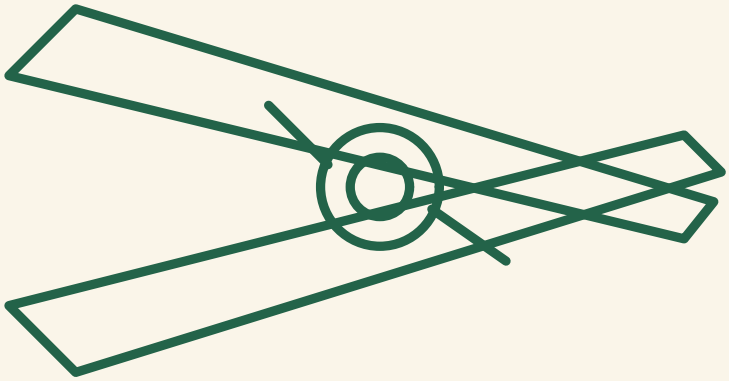
SHUFFLE CONTRAPTION

MATERIALS

18 CARDS / DRAW ONE

BUILD WITH...

Clothespins



Use wooden clothespins as working parts. Try their jaws, spring or wooden halves.

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

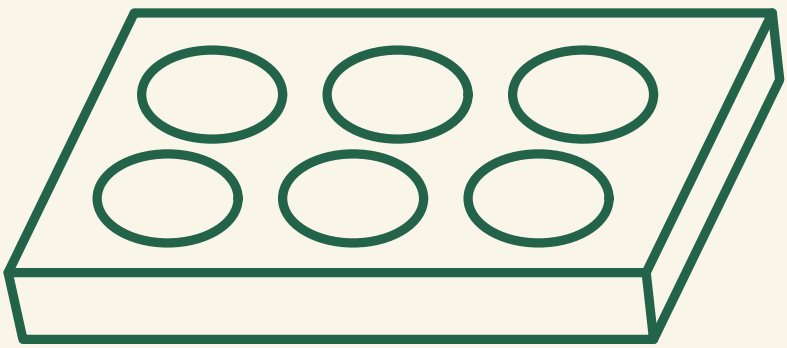
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M14

BUILD WITH...

Egg cartons



Turn a clean egg carton into a useful structure. Explore its cups, ridges and folding lid.

CUP / CUT / CONNECT

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

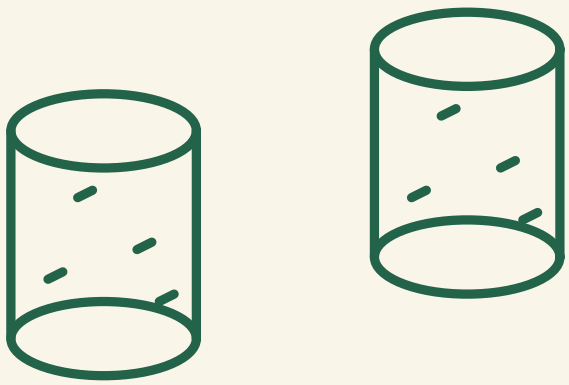
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M15

BUILD WITH...

Corks



Use corks as key parts in your invention. Explore their grip, buoyancy or rounded shape.

FLOAT / ROLL / GRIP

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

SHUFFLE CONTRAPTION

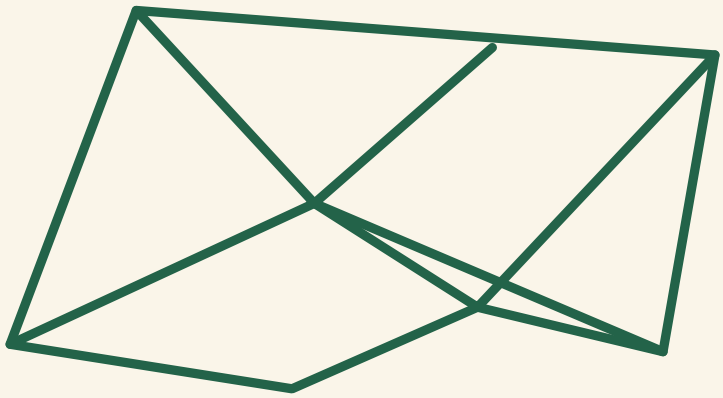
MATERIALS

18 CARDS / DRAW ONE

MATERIALS M16

BUILD WITH...

Aluminum foil



Make foil do useful work. Fold, crumple or layer it into a shape that helps your invention.

FOLD / PRESS / REFLECT

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

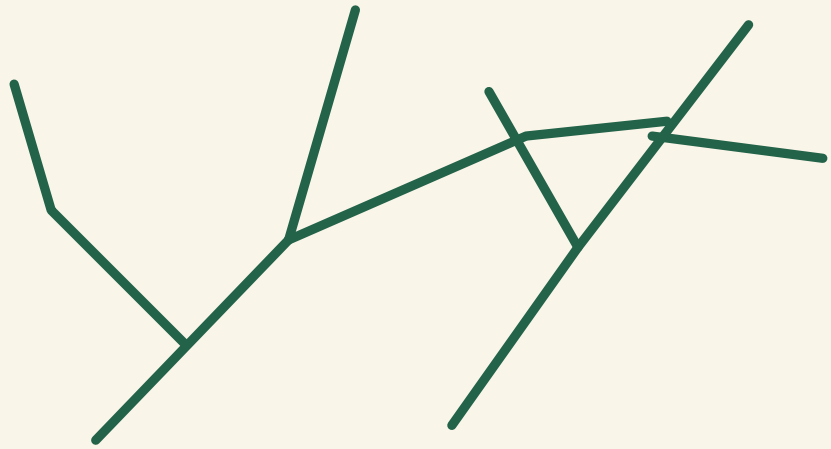
SHUFFLE CONTRAPTION

MATERIALS

18 CARDS / DRAW ONE

BUILD WITH...

Fallen twigs



Gather fallen twigs and build with their natural shapes. Join them into a useful structure.

MATERIALS

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE

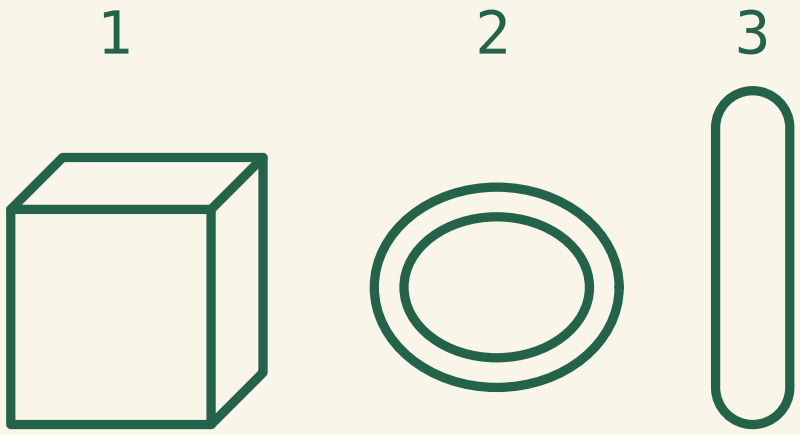
SHUFFLE CONTRAPTION

MATERIALS

18 CARDS / DRAW ONE

BUILD WITH...

Three rescued objects



Choose three clean objects headed for the trash. Give all three a useful role in one contraption.



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

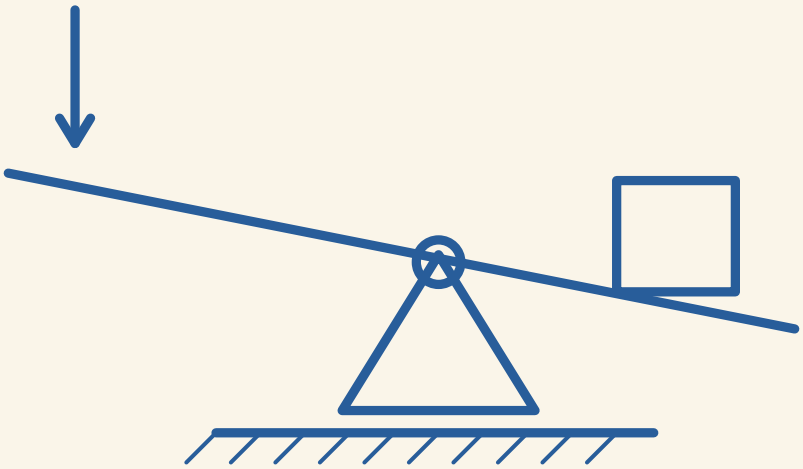
18 CARDS / DRAW ONE



MECHANISM K01

MAKE IT WORK WITH...

Lever



Pivot a bar on a support. Use movement at one end to lift, push or move something at the other.

PIVOT / PUSH / LIFT



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

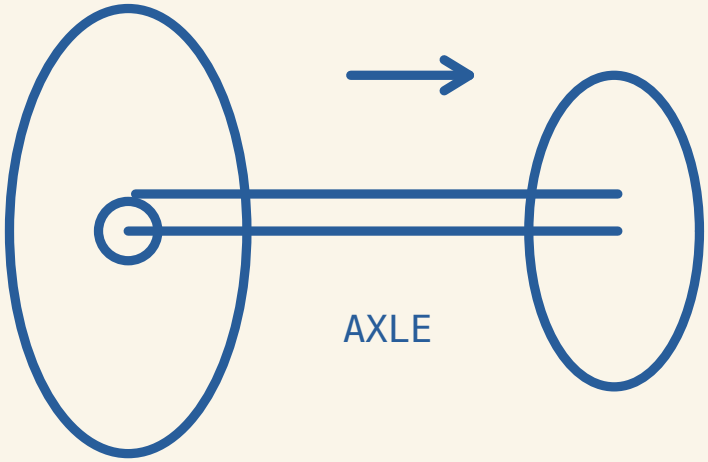
18 CARDS / DRAW ONE



MECHANISM K02

MAKE IT WORK WITH...

Wheel and axle



Connect a wheel to a shaft so they turn together. Use their rotation to help accomplish the mission.

TURN TOGETHER



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

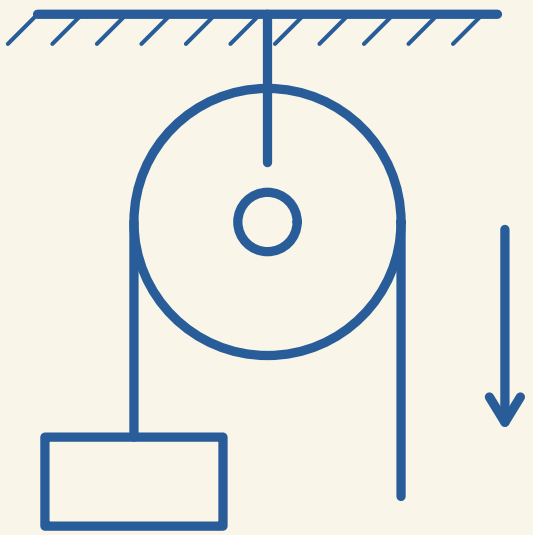
18 CARDS / DRAW ONE



MECHANISM K03

MAKE IT WORK WITH...

Pulley



Run a cord over a turning wheel. Use it to redirect a pull or help move a load.

GUIDE THE PULL



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

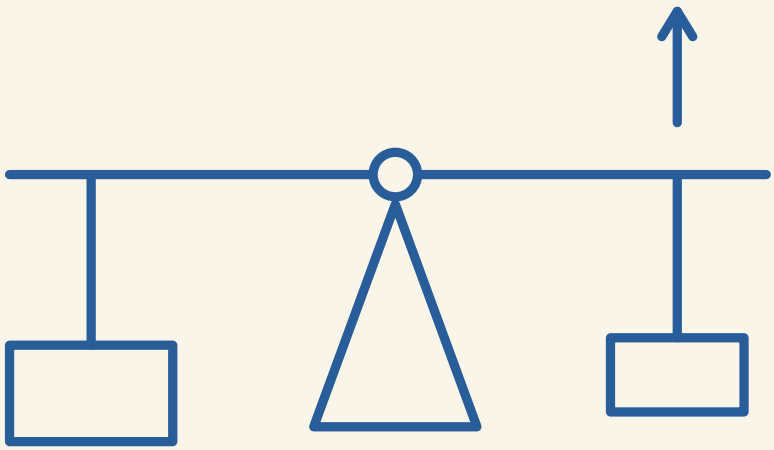
18 CARDS / DRAW ONE



MECHANISM K04

MAKE IT WORK WITH...

Counterweight



Use one weight to balance or move another part. The balancing weight must do useful work.

BALANCE / MOVE



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

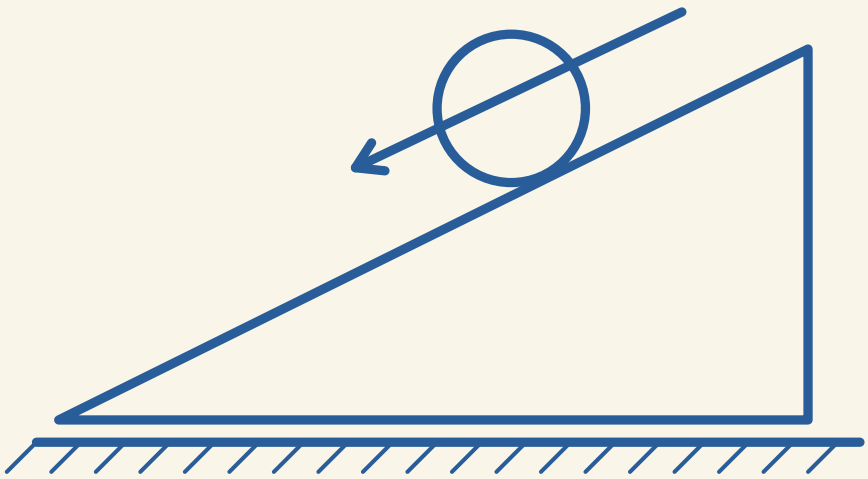
18 CARDS / DRAW ONE



MECHANISM K05

MAKE IT WORK WITH...

Ramp



Use a sloped surface to guide motion or raise an object gradually. Make the slope do useful work.

GUIDE ALONG A SLOPE



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

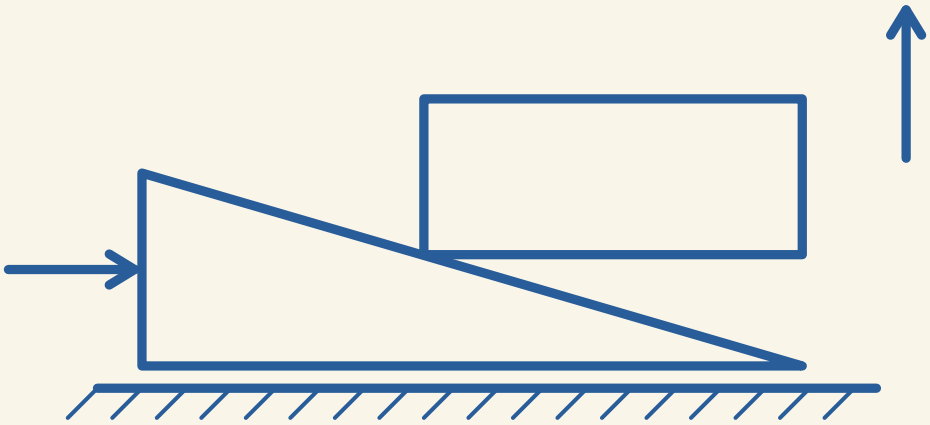
18 CARDS / DRAW ONE



MECHANISM K06

MAKE IT WORK WITH...

Wedge



Push a tapered piece between surfaces. Use its shape to lift, separate or hold something.

PUSH / SEPARATE / HOLD



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

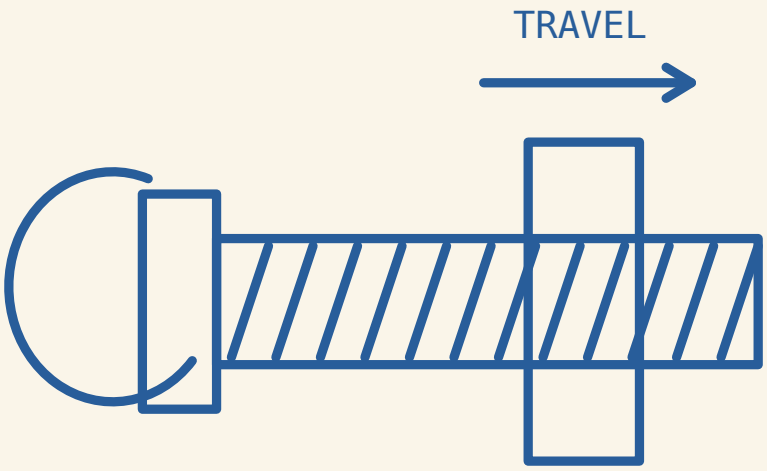
18 CARDS / DRAW ONE



MECHANISM K07

MAKE IT WORK WITH...

Screw



Use a threaded connection to turn rotation into straight-line movement or a holding force.

TURN INTO TRAVEL



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

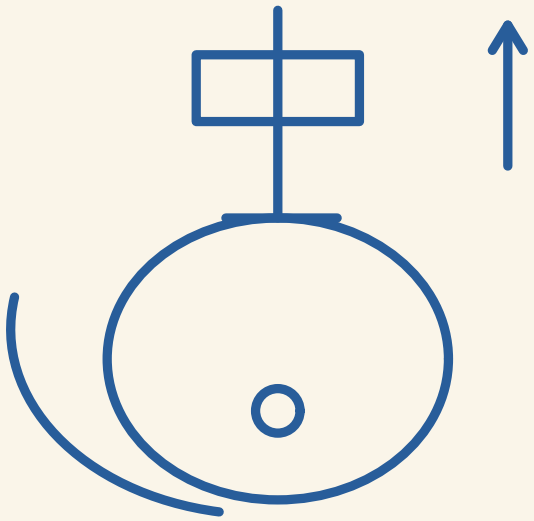
18 CARDS / DRAW ONE



MECHANISM K08

MAKE IT WORK WITH...

Cam and follower



Turn a shaped wheel against a follower. Use its changing profile to lift or move the follower.

TURN INTO A REPEAT



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

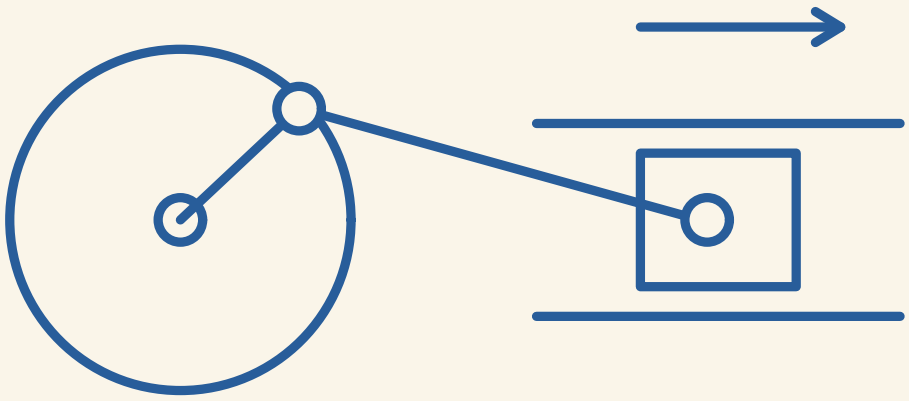
18 CARDS / DRAW ONE



MECHANISM K09

MAKE IT WORK WITH...

Crank and slider



Connect an off-center crank pin to a guided slider. Turn the crank to create back-and-forth travel.

TURN / PUSH / PULL



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

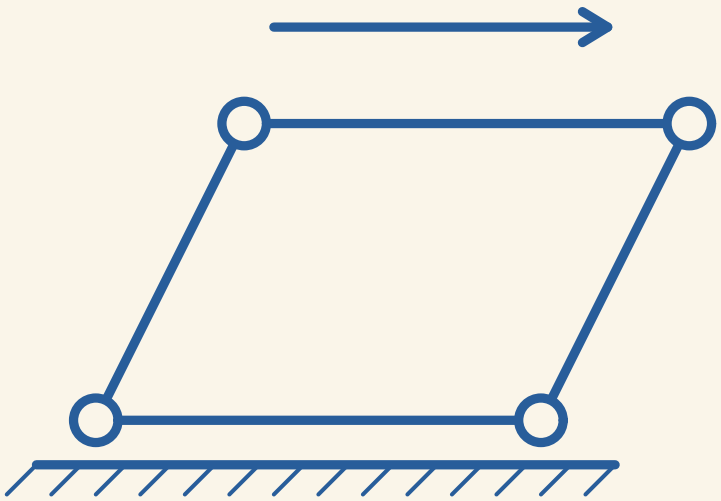
18 CARDS / DRAW ONE



MECHANISM K10

MAKE IT WORK WITH...

Linkage



Join rigid pieces with pivots. Use the connected pieces to guide or transfer movement.

CONNECT THE MOTION



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

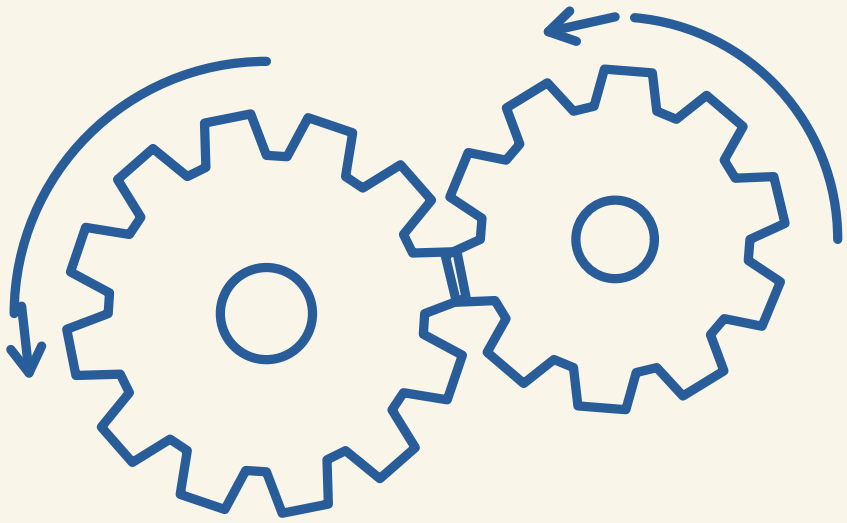
18 CARDS / DRAW ONE



MECHANISM K11

MAKE IT WORK WITH...

Meshing gears



Make toothed wheels engage. Use one gear to turn another and help accomplish the mission.

TEETH TRANSFER MOTION



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

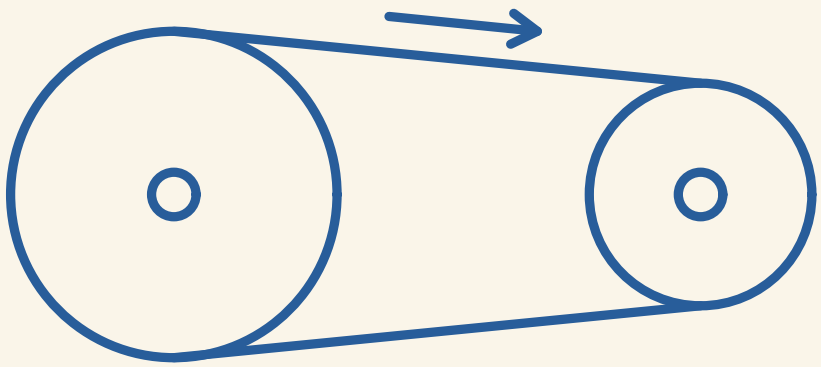
18 CARDS / DRAW ONE



MECHANISM K12

MAKE IT WORK WITH...

Belt drive



Loop a belt around two wheels. Use the belt to carry rotation from one wheel to the other.

CONNECT TWO WHEELS



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

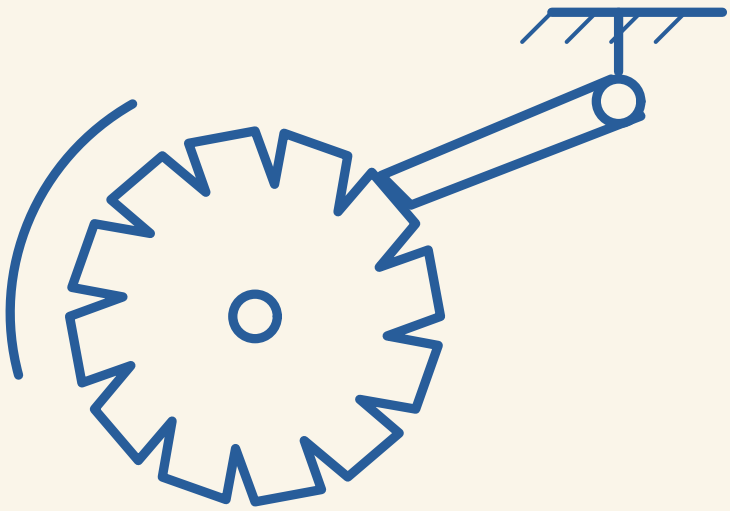
18 CARDS / DRAW ONE



MECHANISM K13

MAKE IT WORK WITH...

Ratchet and pawl



Let a catch engage a toothed wheel or strip. Allow motion one way and resist it the other way.

ADVANCE / HOLD



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

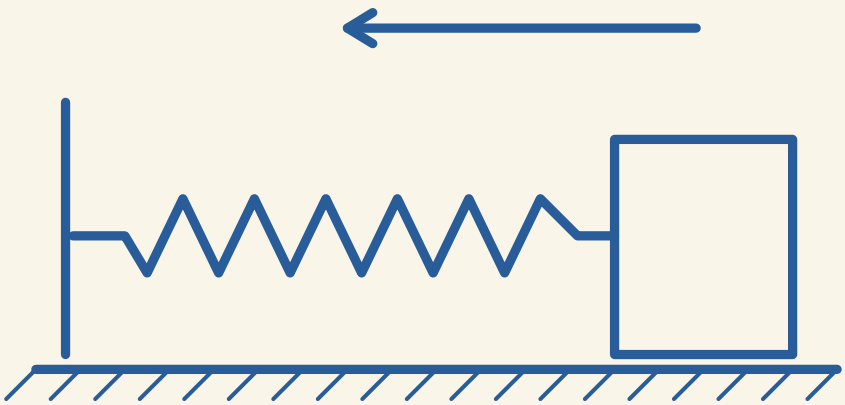
18 CARDS / DRAW ONE



MECHANISM K14

MAKE IT WORK WITH...

Spring return



Store energy in a spring or elastic part. Release it to bring a moving part back toward its start.

STRETCH / RELEASE



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

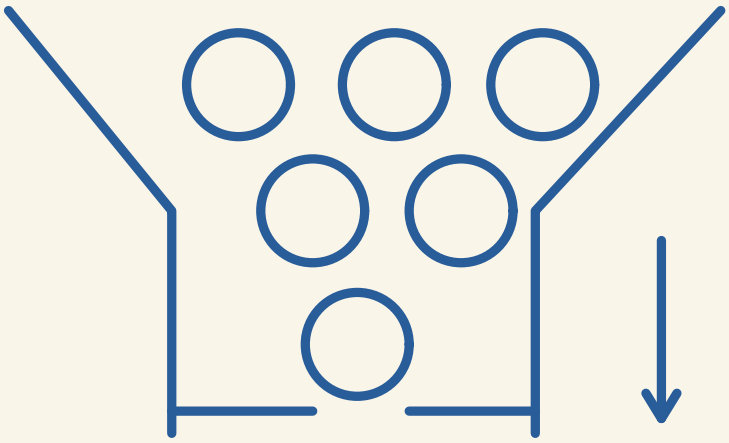
18 CARDS / DRAW ONE



MECHANISM K15

MAKE IT WORK WITH...

Gravity feed



Arrange parts so gravity supplies the motion. Guide objects toward a useful destination.

LET GRAVITY WORK



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

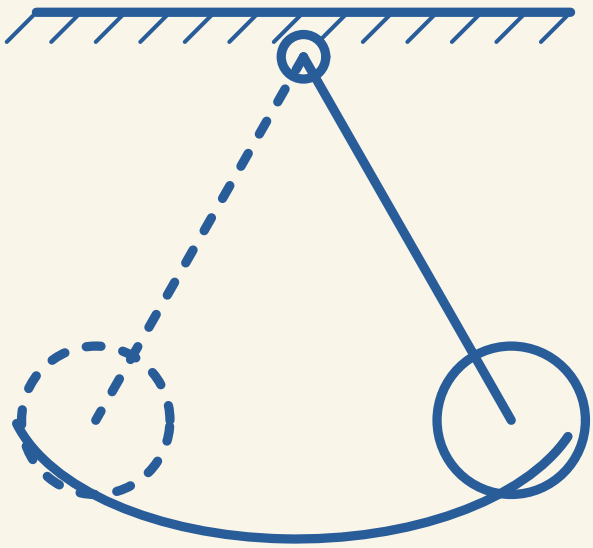
18 CARDS / DRAW ONE



MECHANISM K16

MAKE IT WORK WITH...

Pendulum



Hang a weight from a pivot and let it swing. Use the repeated motion to help with your mission.

SWING FROM A PIVOT



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

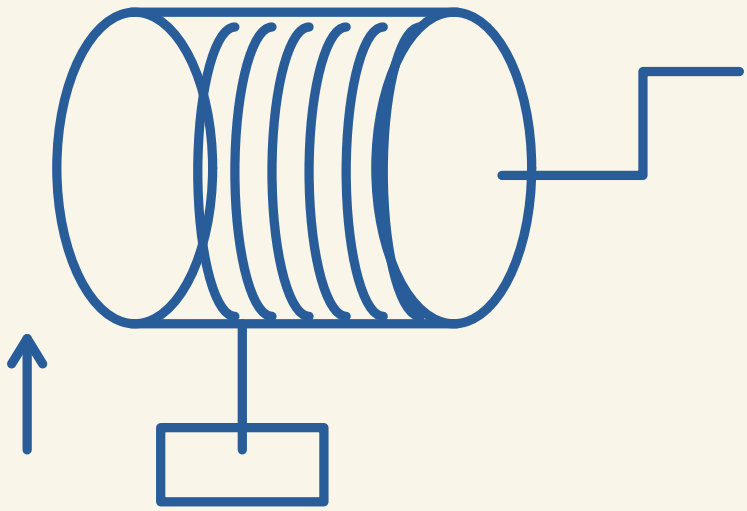
18 CARDS / DRAW ONE



MECHANISM K17

MAKE IT WORK WITH...

Spool and cord



Wind or unwind a cord around a drum. Use the changing cord length to move another part.

WIND / LIFT / RELEASE



MECHANISM

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

MECHANISM

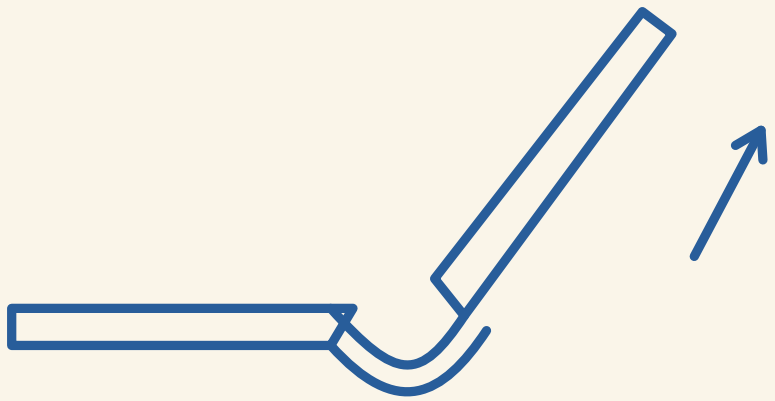
18 CARDS / DRAW ONE



MECHANISM K18

MAKE IT WORK WITH...

Flexible hinge



FLEX

Join two parts with a bendable strip. Make movement happen by bending the joint.

BEND AT THE JOINT



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

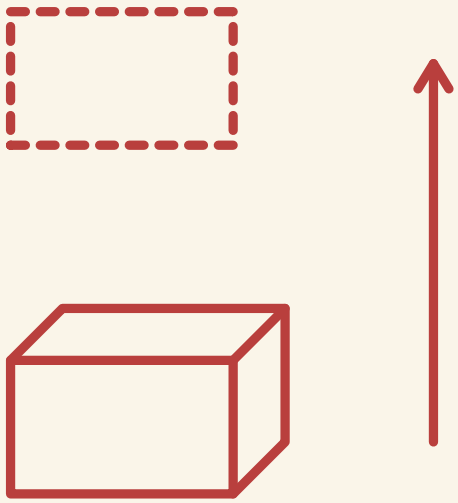
MISSION



MISSION G01

MAKE IT...

Lift a small object



Raise a small object from a lower level to a higher one. Let your mechanism do the lifting.

TEST: SHOW A CLEAR LIFT



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

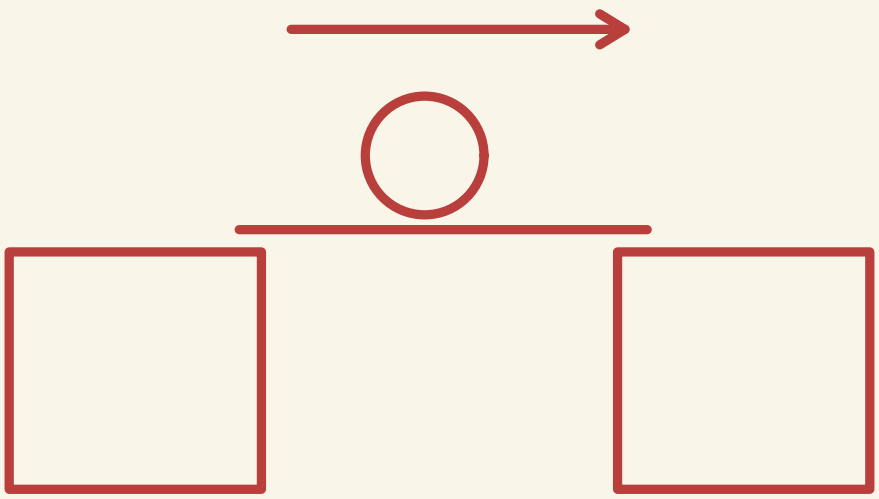
MISSION



MISSION G02

MAKE IT...

Cross a gap



Move a small object from one side of a gap to the other. Keep it out of the space below.

TEST: REACH THE FAR SIDE



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

MISSION



MISSION G03

MAKE IT...

Deliver a message



Carry a written or drawn message from its starting point to a waiting recipient.

TEST: MESSAGE RECEIVED



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

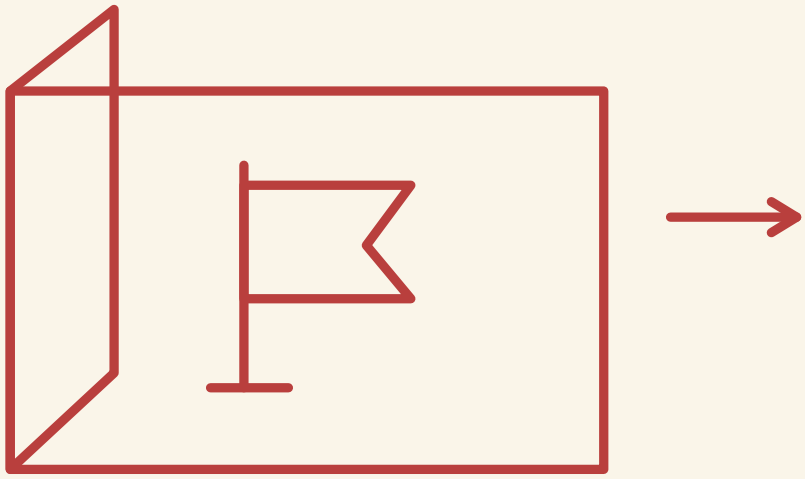
MISSION



MISSION G04

MAKE IT...

Reveal something hidden



Keep an object or message concealed. Operate your contraption to bring it into view.

TEST: HIDDEN TO VISIBLE



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

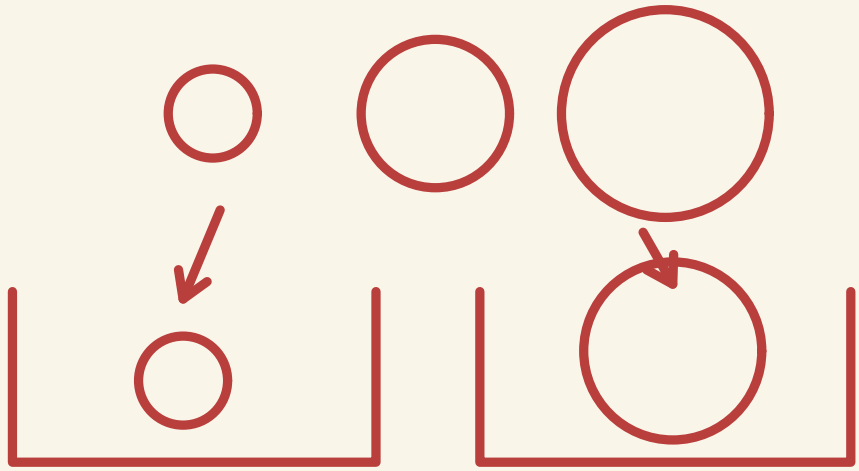
MISSION



MISSION G05

MAKE IT...

Sort by size



Separate a mixed collection into at least two size groups. Make the contraption help decide.

TEST: TWO DISTINCT GROUPS



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

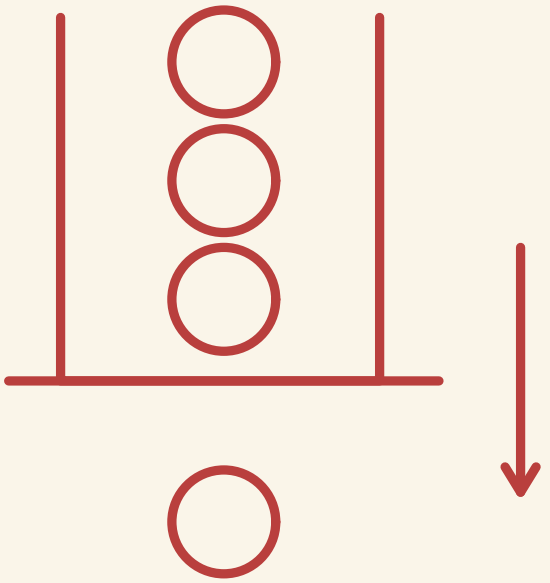
MISSION



MISSION G06

MAKE IT...

Dispense one at a time



Hold several similar objects. Release just one during each operating cycle.

TEST: ONE PER CYCLE



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

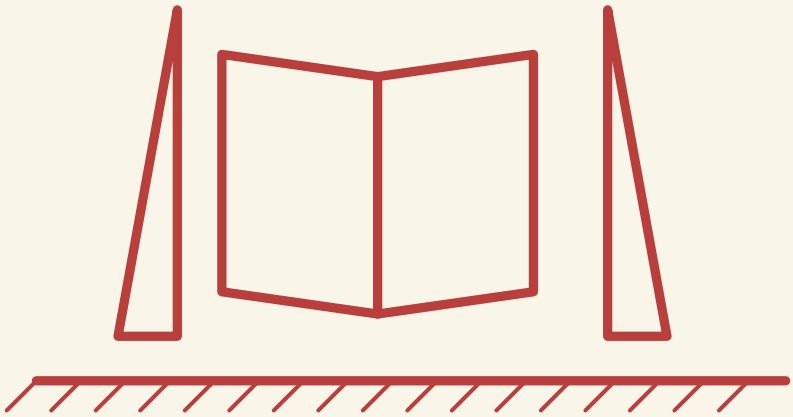
MISSION



MISSION G07

MAKE IT...

Hold it in position



Keep an object in a chosen position without holding it yourself. Show that it stays put.

TEST: LET GO; IT STAYS



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

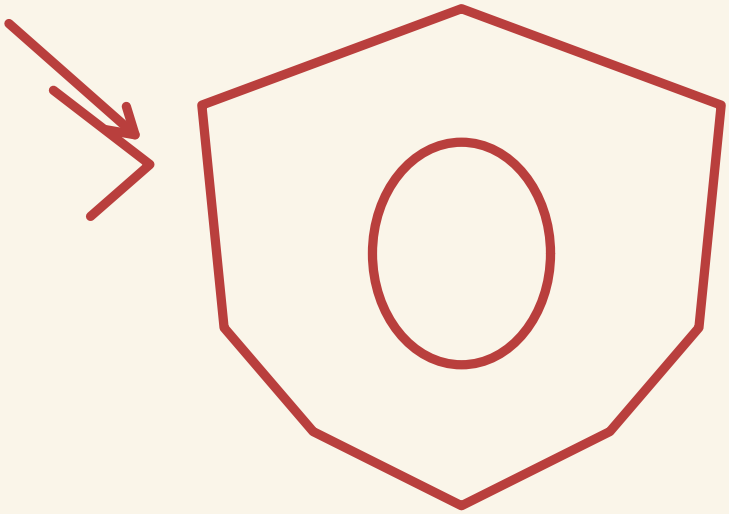
MISSION



MISSION G08

MAKE IT...

Protect something fragile



Build a way to protect a fragile object from a small bump. Use a harmless stand-in first.

TEST: SURVIVE A SMALL BUMP



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

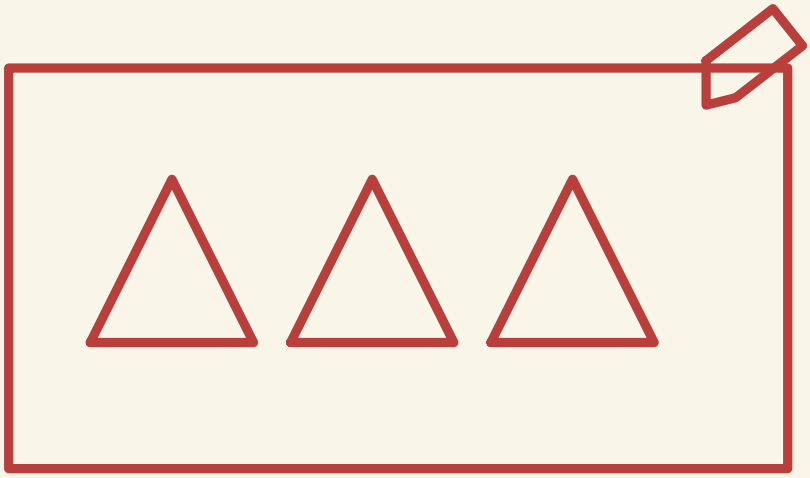
MISSION



MISSION G09

MAKE IT...

Draw a repeating pattern



Move a drawing tool or a surface to make a pattern repeat. Let the mechanism guide it.

TEST: REPEAT THREE TIMES



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

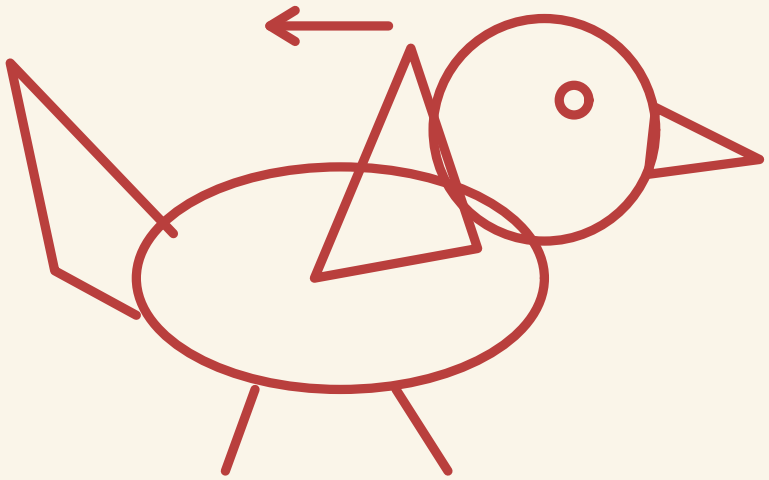
MISSION



MISSION G10

MAKE IT...

Bring a creature to life



Give an imagined creature a convincing movement. Make it nod, flap, crawl or express a mood.

TEST: SHOW ITS MOVEMENT



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

MISSION



MISSION G11

MAKE IT...

Give a visible signal



Make a clear visual change that someone can recognize from across a table.

TEST: SPOT THE CHANGE



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

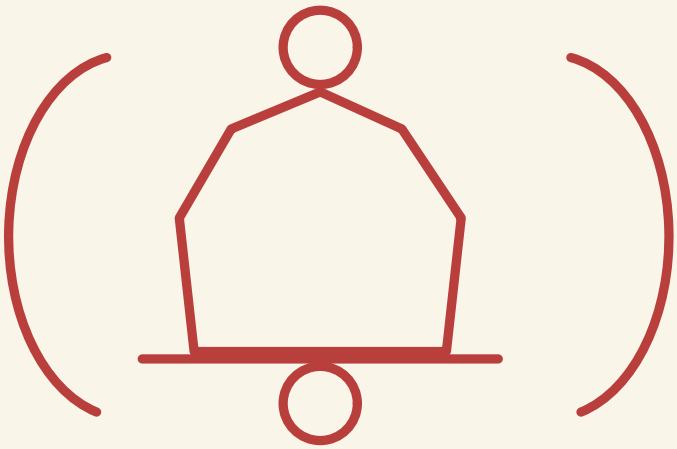
MISSION



MISSION G12

MAKE IT...

Make a sound



Create an intentional sound when your contraption operates. Make it happen again on demand.

TEST: SOUND ON DEMAND



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

MISSION



MISSION G13

MAKE IT...

Compare two weights



Show which of two small objects is heavier, or whether they balance. Use the same test for both.

TEST: SWAP AND COMPARE



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

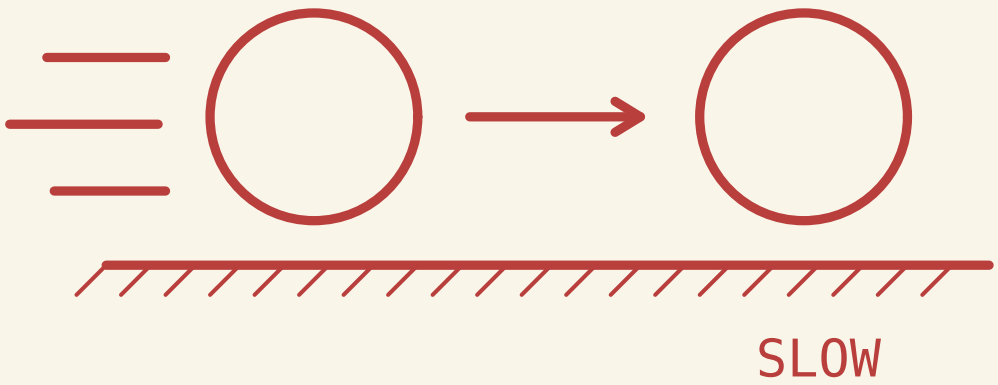
MISSION



MISSION G14

MAKE IT...

Slow a moving object



Make an object move more slowly than it would without your contraption. Compare both runs.

TEST: COMPARE THE MOTION



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

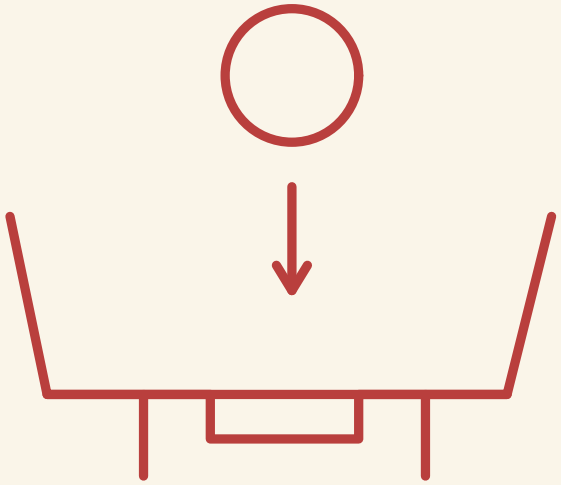
MISSION



MISSION G15

MAKE IT...

Catch and release



Catch a small moving object, hold it briefly, then release it through your mechanism.

TEST: CATCH / HOLD / LET GO



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

MISSION



MISSION G16

MAKE IT...

Fold into less space



Make a useful structure that folds into a smaller form. Open it again and show it still works.

TEST: FOLD AND REOPEN



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

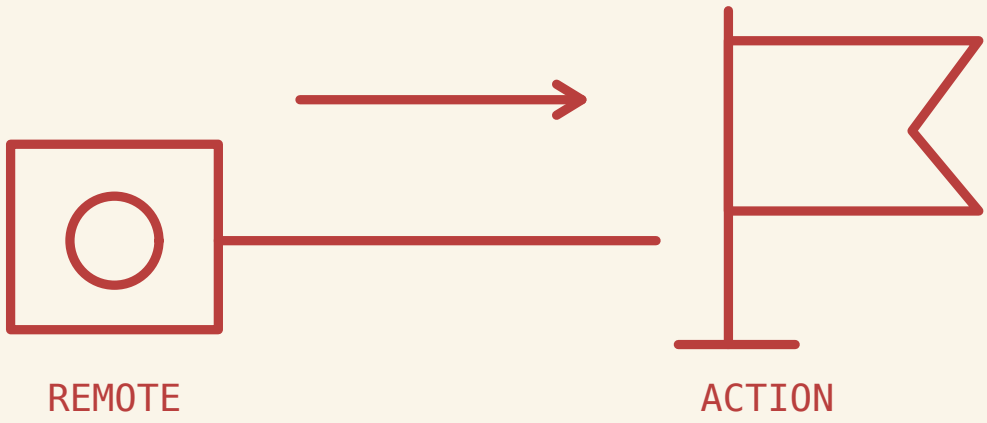
MISSION



MISSION G17

MAKE IT...

Operate from a distance



Control an action from beyond arm's reach. Keep your hands away from the part doing the job.

TEST: WORK IT FROM AFAR



MISSION

18 CARDS / DRAW ONE

CONTRAPTION SHUFFLE



CONTRAPTION SHUFFLE

18 CARDS / DRAW ONE

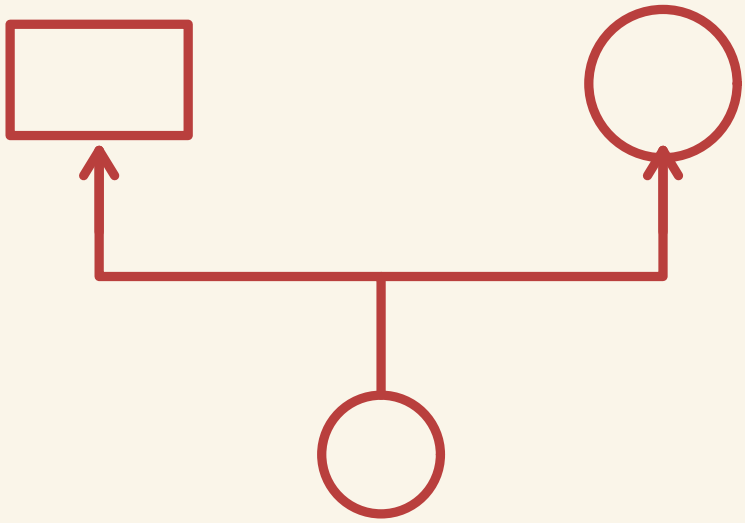
MISSION



MISSION G18

MAKE IT...

Move two things together



Use one input to make two separate parts move at the same time. Give both parts a visible job.

TEST: ONE INPUT; TWO MOVES